

Session 10 Quiz - Strengths & Limits of Generative AI

Track A | Grades 5-8

Q1 [Type: MCQ | Difficulty: Easy | QID:57049]

Fill in the blank: AI is great at _____, which provides endless ideas when you are stuck and need inspiration.

- A. fact-checking
- B. physical labor
- C. brainstorming
- D. emotional empathy

Correct Answer: C. brainstorming

Explanation: Generating a high volume of ideas quickly is one of the primary superpowers of generative AI mentioned in the material. Brainstorming helps when you are stuck and need creative options to choose from.

Q2 [Type: MCQ | Difficulty: Easy | QID:57050]

Complete the following sentence: When an AI confidently makes up information that is not true, it is known as a _____.

- A. hallucination
- B. system update
- C. draft
- D. pattern match

Correct Answer: A. hallucination

Explanation: This term refers to instances where AI generates false or sketchy information with a high degree of confidence. Hallucinations are a key limitation of generative AI.

Q3 [Type: MCQ | Difficulty: Easy-Medium | QID:57051]

Fill in the blank: The _____ concept means that the human stays in charge while the AI does the fast, tireless work.

- A. Machine-First
- B. Human-in-the-Loop
- C. AI-Only
- D. Autopilot-Exclusive

Correct Answer: B. Human-in-the-Loop

Explanation: This core idea emphasizes that the best results come from combining AI's speed with human wisdom and decision making. The human stays in charge and remains an essential part of the process.

Q4 [Type: MCQ | Difficulty: Easy | QID:57052]

Complete the sentence: AI is compared to a _____ because it is a powerful tool, but it is only effective for the right jobs.

- A. lightbulb
- B. textbook
- C. calculator
- D. hammer

Correct Answer: D. hammer

Explanation: The hammer analogy illustrates that even a super powered tool should not be used for everything, like hugging a dog. AI is powerful but only for the right tasks.

Q5 [Type: MCQ | Difficulty: Easy | QID:57053]

Fill in the blank: AI doesn't actually 'think' like a person; instead, it uses _____ to predict what should come next.

- A. memory
- B. magic
- C. pattern-matching
- D. telepathy

Correct Answer: C. pattern-matching

Explanation: AI functions by finding patterns in its training data rather than possessing real understanding or common sense. It predicts the next word or action based on learned patterns.

Q6 [Type: True/False | Difficulty: Easy | QID:57054]

True or False: AI can reliably tell you what the President said in a speech that happened yesterday.

- A. True
- B. False

Correct Answer: B. False

Explanation: AI often has outdated information and may hallucinate details about very recent events. This is one of the weaknesses mentioned regarding outdated information.

Q7 [Type: True/False | Difficulty: Easy | QID:57055]

True or False: Using AI to explain gravity in different ways, such as using a magnet or a trampoline analogy, is a good use of its strengths.

A. False

B. True

Correct Answer: B. True

Explanation: One of AI's superpowers is its ability to re-explain concepts multiple ways until they click for the user. This is a strength example from the material.

Q8 [Type: True/False | Difficulty: Easy-Medium | QID:57056]

True or False: AI has enough common sense to know exactly when a joke might be inappropriate for your grandmother.

A. True

B. False

Correct Answer: B. False

Explanation: AI lacks real common sense and often misses obvious social nuances that humans understand naturally. The example about frying an egg on the sun shows this limitation.

Q9 [Type: True/False | Difficulty: Easy-Medium | QID:57057]

True or False: If AI does your math homework for you, you are using the tool correctly to help you learn.

A. False

B. True

Correct Answer: A. False

Explanation: Letting AI do the work for you prevents you from learning. You should use it to explain concepts instead. The goal is understanding, not just finishing the task.

Q10 [Type: True/False | Difficulty: Easy | QID:57058]

True or False: AI is a reliable source for serious medical advice during a health emergency.

A. True

B. False

Correct Answer: B. False

Explanation: AI is not a doctor, and serious or risky situations require experts, not AI guesses. This is covered in the section When Should You NOT Use AI.

Q11 [Type: MCQ | Difficulty: Easy-Medium | QID:57059]

Which of the following is considered one of AI's 'superpowers'?

- A. Verifying that every fact it provides is 100% true
- B. Doing in 10 seconds what takes a human 1 hour
- C. Predicting exactly what your mom said this morning
- D. Understanding the world exactly like a 5-year-old human

Correct Answer: B. Doing in 10 seconds what takes a human 1 hour

Explanation: Speed and efficiency in processing information is a major strength of generative AI. AI cannot verify every fact, predict specific private conversations, or truly understand the world like a human.

Q12 [Type: MCQ | Difficulty: Easy-Medium | QID:57061]

Why does the lesson suggest using Google instead of AI for news about a game played last night?

- A. Because AI doesn't know how to speak English
- B. Because AI is only allowed to be used for schoolwork
- C. Because AI information can be outdated
- D. Because Google is more creative than AI

Correct Answer: C. Because AI information can be outdated

Explanation: AI training data often ends at a certain point, making it unaware of events that happened very recently. Google can access live, current news.

Q13 [Type: MCQ | Difficulty: Easy-Medium | QID:57062]

In the pilot story analogy, what is the role of the AI 'autopilot'?

- A. To help fly the plane while the pilot watches
- B. To replace the need for a human pilot entirely
- C. To take full responsibility if the plane crashes
- D. To decide where the plane is going

Correct Answer: A. To help fly the plane while the pilot watches

Explanation: The autopilot assists with the work, but the human pilot remains responsible and in control. This illustrates the Human-in-the-Loop concept.

Q14 [Type: MCQ | Difficulty: Medium | QID:57063]

When planning a class assembly using the 'Human-in-the-Loop' method, what is a task only the HUMAN should do?

- A. Write 5 different rough drafts for a speech
- B. Translate a kindness note into 10 different languages
- C. Choose the ideas that fit the school's budget and culture
- D. Generate 10 creative ideas in 5 seconds

Correct Answer: C. Choose the ideas that fit the school's budget and culture

Explanation: Humans apply judgment, context, and knowledge of constraints that AI does not understand. The other tasks are strengths of AI (drafting, translation, brainstorming).

Q15 [Type: MCQ | Difficulty: Easy-Medium | QID:57064]

Which of these is a valid reason to NOT use AI for information about real people?

- A. AI is too slow to look up famous people
- B. Real people are too complex for AI to understand at all
- C. AI is only designed to talk about fictional characters
- D. AI can fabricate or make up details about individuals

Correct Answer: D. AI can fabricate or make up details about individuals

Explanation: AI often hallucinates facts about people, which can lead to spreading misinformation. This is a key limitation when dealing with real individuals.

Q16 [Type: MCQ | Difficulty: Easy-Medium | QID:57066]

If an AI tells you that you can fry an egg on the sun, it is showing a lack of _____.

- A. common sense
- B. speed
- C. creativity
- D. vocabulary

Correct Answer: A. common sense

Explanation: AI follows physics patterns in data but does not realize the silliness or obvious impossibility of certain ideas. Common sense is a human trait that AI lacks.

Q17 [Type: MCQ | Difficulty: Easy-Medium | QID:57067]

What is the purpose of performing an 'AI Tool Analysis'?

- A. To find out which AI tool is the most expensive
- B. To prove that AI is better than a human pilot
- C. To learn how to code your own AI from scratch
- D. To create a personal guide for using AI wisely

Correct Answer: D. To create a personal guide for using AI wisely

Explanation: Analyzing a tool's strengths and weaknesses helps you decide when to use it and when to avoid it. This creates a personal guide for wise use.

Q18 [Type: MCQ | Difficulty: Easy-Medium | QID:57068]

Which task represents AI at its BEST?

- A. Giving you a starting point to build from
- B. Reporting on a live emergency in your neighborhood
- C. Correcting its own biased training data
- D. Making final decisions for a serious situation

Correct Answer: A. Giving you a starting point to build from

Explanation: AI is excellent at creating drafts that humans can then edit and improve. It is not good for live emergencies, cannot easily correct its own bias, and should not make final serious decisions.

Q19 [Type: MCQ | Difficulty: Easy-Medium | QID:57070]

Why is it important to double-check 'facts that must be right' when using AI?

- A. AI can sound confident even when it is wrong
- B. Checking facts makes the AI smarter over time
- C. AI only works correctly for math, not for other subjects
- D. AI will charge you more money if it gets a fact wrong

Correct Answer: A. AI can sound confident even when it is wrong

Explanation: The hallucination problem means AI provides false information with the same tone as true information. Confidence does not equal accuracy, so double-checking is essential.

Q20 [Type: MCQ | Difficulty: Easy-Medium | QID:57071]

What is the main takeaway of the 'Pilot Story' regarding accountability?

- A. The AI is responsible if the pilot falls asleep
- B. No one is responsible if the AI makes a mistake
- C. The passengers are responsible for checking the autopilot
- D. The human pilot is always responsible for the flight

Correct Answer: D. The human pilot is always responsible for the flight

Explanation: Responsibility for the outcome never transfers from the human to the machine. The human stays accountable for the final decisions and results.